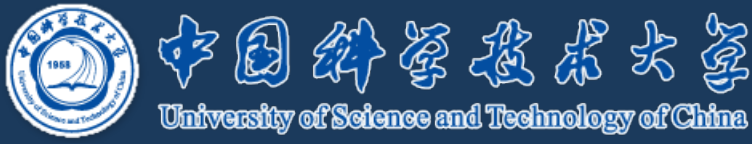


M-DaQ: Retrieving Samples with Multilingual Diversity and Quality for Instruction Fine-Tuning Datasets

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1 MOTIVATION

- Multilingual IFT data is scarce, skewed toward English, and lacks systematic curation
- Llama-3 IFT dataset: only 3.01% multilingual samples
- No language-agnostic quality scoring for multilingual data
- Data diversity studied only in English settings
- Superficial Alignment Hypothesis (SAH) unverified in multilingual contexts

Three Challenges Addressed:

- C1** No extensible quality scoring for multilingual IFT
→ **QSM with triplet loss**
- C2** Diversity selection limited to English
→ **DAS inspired by MMR**
- C3** SAH unverified in multilingual settings
→ **1K → 52K study, 8 languages**

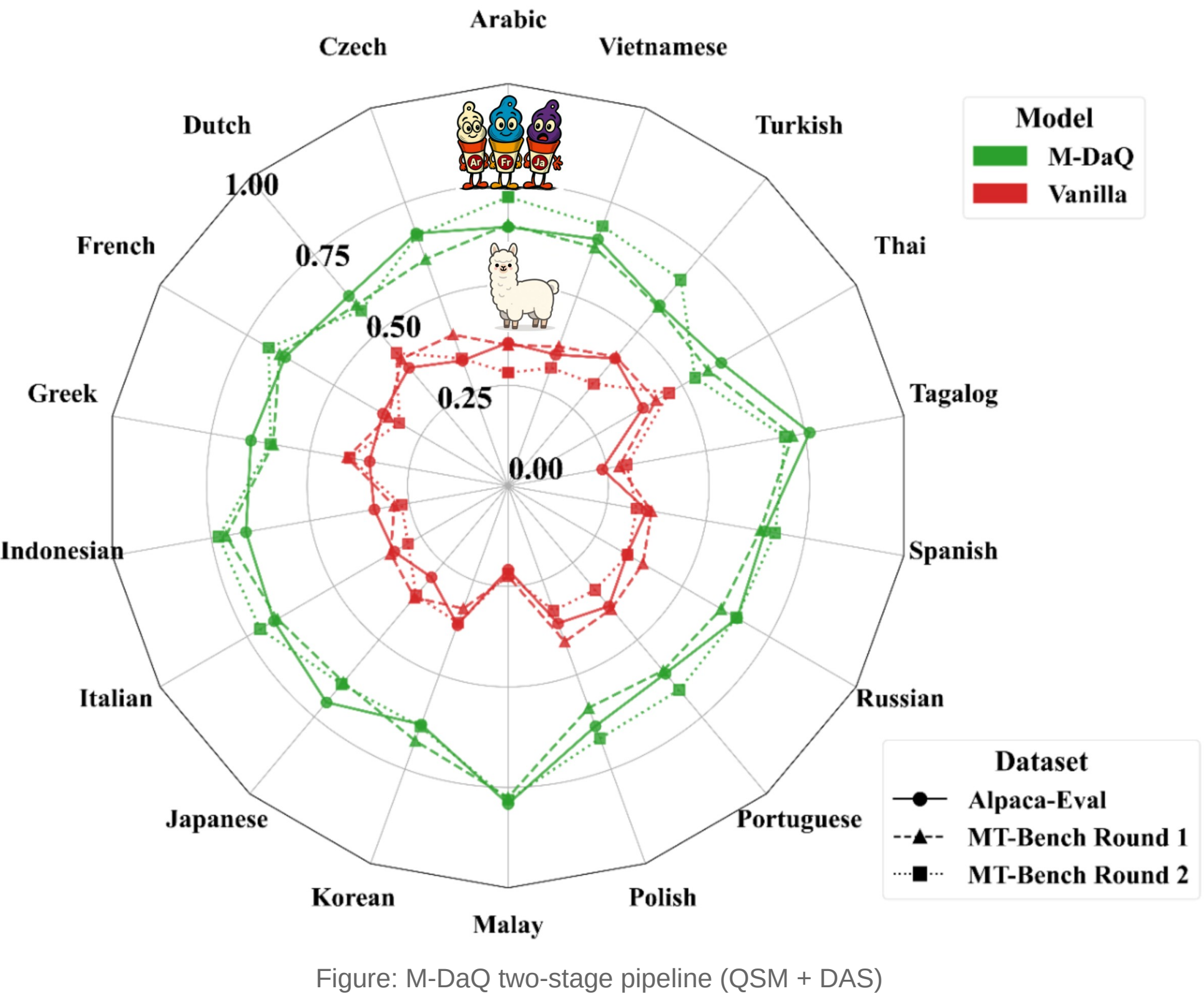
2 METHOD: M-DaQ FRAMEWORK

Stage 1: Quality Scoring Model (QSM)

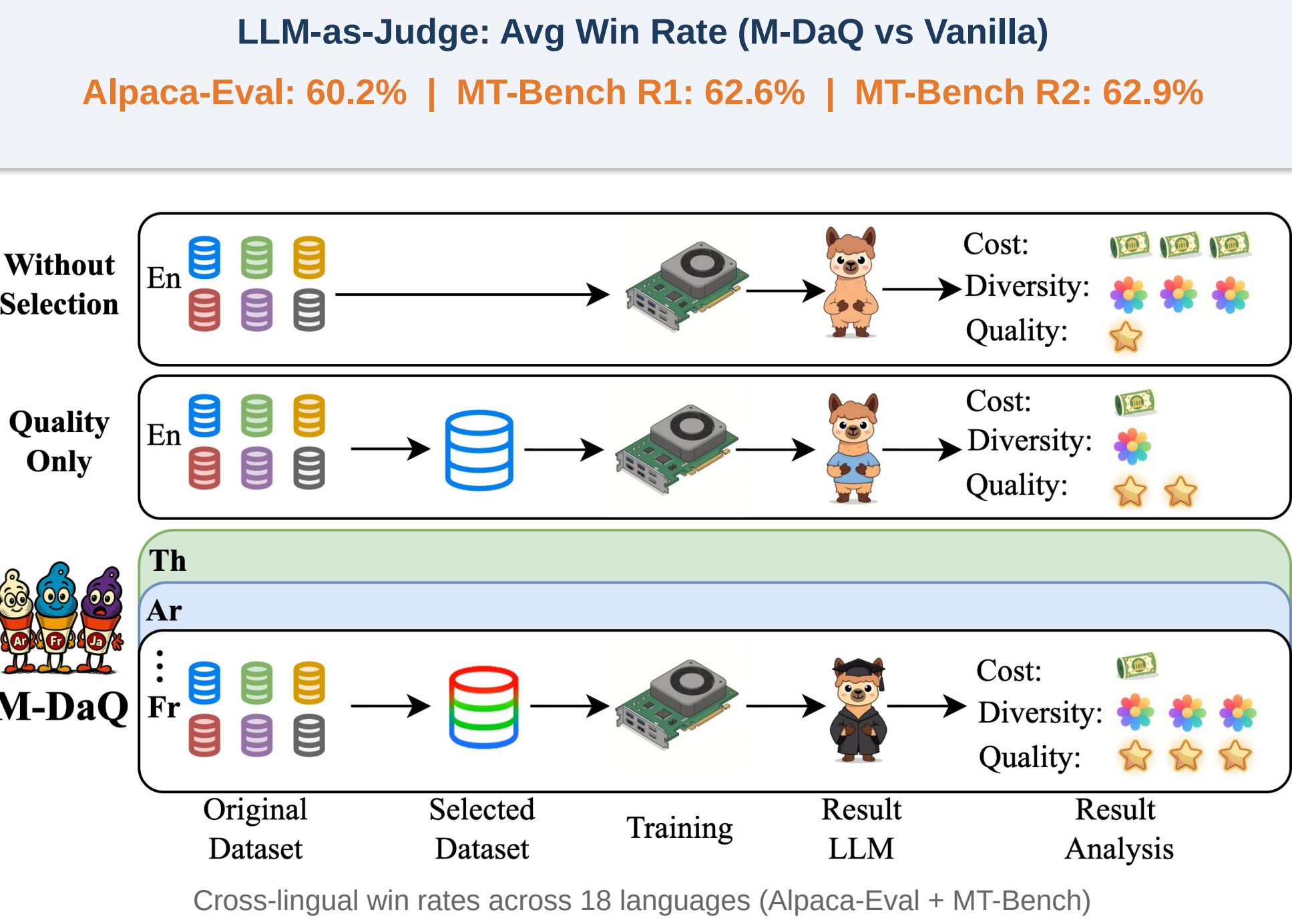
- Fine-tuned on ~2.3K expert-revised samples × 18 languages (from MIDB)
- Triplet loss: instruction → positive (expert-revised) vs negative (original/MT)
- Language-agnostic quality signal from embedding space

Stage 2: Diversity-Aware Selection (DAS)

- MMR-inspired: Quality ≈ Relevance, Diversity ≈ Novelty
- Two-stage pipeline reduces $O(n^2)$ → $O(n \log n)$
 - Stage A: Select top-n by QSM score (quality subset)
 - Stage B: Greedily augment from uncovered clusters (diversity subset)
- Ratio $n_{\text{quality}} : n_{\text{diversity}} = 6:1$ (empirically tuned)



3 KEY RESULTS



Key Observations:

- Consistent improvement across all 18 languages
- Gains MORE pronounced in low-resource languages (Tagalog, Malay)
- M-DaQ effectively mitigates linguistic imbalance and data scarcity

Human Evaluation (6 Langs, 900 Samples, 58 Person-Hours):

Language	Alpaca-Eval	MT-Bench R1	MT-Bench R2
Japanese	86%	84%	80%
Korean	94%	94%	94%
Russian	70%	86%	84%
Portuguese	80%	78%	84%
Greek	58%	76%	82%
French	80%	92%	88%
Average	78.0%	85.0%	85.3%

Cultural Localization Example:

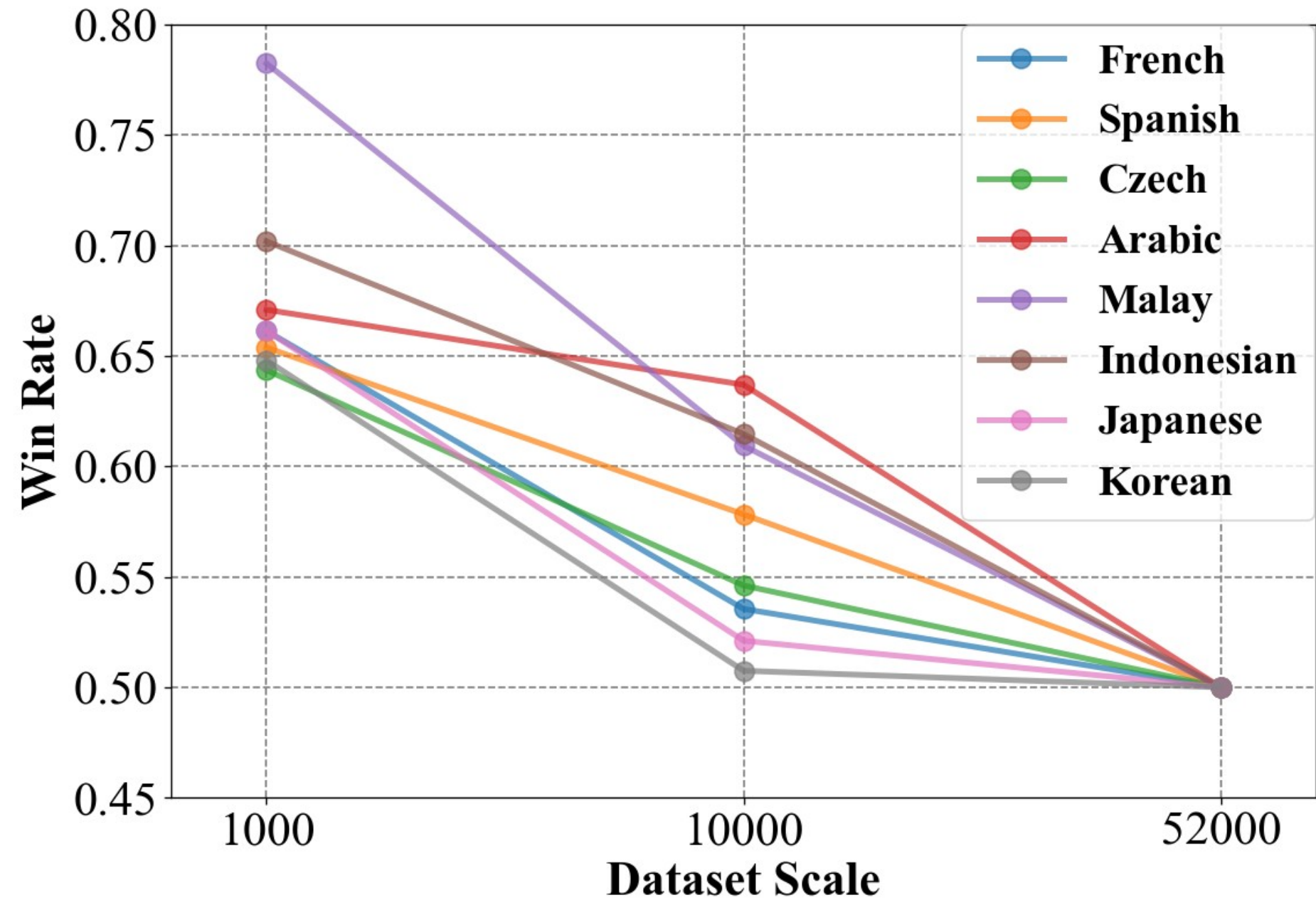
Instruction:
Rédigez un blog de voyage captivant sur un voyage récent en Corse, en mettant en évidence les expériences culturelles et les attractions incontournables. (Write a captivating travel blog about a recent trip to Corsica, highlighting cultural experiences and must-see attractions.)

Response:
Faire un voyage en Corse est l'une des aventures ...**(Making a trip to Corsica is one of the most fascinating and amazing adventures...)**
L'une des expériences les plus mémorables de mon voyage a été mon visit à **Piana (a historic mountain village in southern Corsica)**, un petit village perdu dans les montagnes de l'intérieur de l'île. Le village est fièrement animé par sa population de langue **corse (Corsican language)** et sa cuisine **traditionnelle (Corsican traditional cuisine reflects centuries of pastoral life, Mediterranean influences)**. J'ai eu la chance de goûter certains des plats locaux, y compris la ravioli et la **figatelli (a traditional Corsican dried pork sausage)**, et de les apprécier avec une coupe de vin rouge **corsique (Corsican wines)**.

French travel query about Corsica → M-DaQ references Piana, figatelli, local cultural conventions

4 SAH IN MULTILINGUAL SETTINGS

First systematic investigation of the Superficial Alignment Hypothesis across languages.



Key Findings:

- Diminishing returns: 1K curated > 52K unfiltered
 - SAH holds multilingually
- 10K → −10.1% avg win rate vs 1K baseline
- 52K → −6.2% avg win rate vs 1K baseline
- Language-dependent sensitivity:
 - Arabic shows flatter curve than French
 - (lower pretraining readiness)
- A few thousand high-quality samples suffice
 - for effective multilingual alignment

5 CONCLUSION

- M-DaQ = QSM (quality) + DAS (diversity):
 - compact, high-fidelity multilingual IFT subsets
- 60%+ avg win rate across 18 languages
 - on Alpaca-Eval & MT-Bench
- Human eval confirms gains in cultural relevance,
 - contextual appropriateness, instruction-following
- First empirical validation of SAH
 - in multilingual settings
- Code released: github.com/zhaocorey/M-DaQ

Code & Resources

GitHub: github.com/zhaocorey/M-DaQ

Paper

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arXiv: 2509.15549

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